Code No: 138AC **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year II Semester Examinations, July/August - 2022
ADVANCED ALGORITHMS
(Computer Science and Engineering)

Time: 3 Hours **Max.Marks:75**

Answer any five questions
All questions carry equal marks
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1.a) Explain the algorithm of quick sort with an example.
b) Write an algorithm of Longest Common Subsequence. [7+8]

2.a) Write an algorithm of Optimal Binary Search Tree.
b) Sort the following list of elements by using randomized heap sort. [8+7]
54, 89, 20, 77, 10, 46, 92, 36

3.a) Write an algorithm of Kruskal's minimum cost spanning tree.
b) Explain about Amortized Analysis. [8+7]

4.a) Explain about the activity selection problem with an example.
b) Write an algorithm of Max Flow problem. [7+8]

5.a) Derive the time complexity of Strassen's Matrix multiplication.
b) Explain about the Zero – one principle. [8+7]

6. Explain about the merging networks with an example. [15]

7.a) Apply the Rabin - karp algorithm for the following text and pattern and Illustrate the intermediate steps
text = 314593671028
pattern = 67
b) Write an algorithm of Naïve String Pattern and also find the time complexity of the same. [7+8]

8.a) Explain about the NP- Complete and also explain any one NP-Complete problem with an example.
b) Explain about the Travelling Sales Person problem by using an approximation algorithm. [7+8]

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